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Challenges and future directions

Throughout this monograph, we have provided a comprehensive re-interpretation of GNN-based recommender systems from a topological perspective. By modeling the user-item interaction space as a bipartite graph, we have moved beyond treating GNNs as black-box predictors and instead have shown how their performance is fundamentally linked to the topological characteristics of the data. Our explanatory framework has demonstrated that these properties are not just passively present but are actively, if sometimes implicitly, encoded within the models' formulations.

This deep dive into the interplay between graph topology and recommendation performance serves as a natural springboard into the next crucial discussion: the challenges and future directions for the field. The very success of GNNs in this domain opens up a new frontier of complex questions. *If topology is so critical, how can we design models that exploit it more deliberately? If our models are sensitive to these properties, what are the implications of noisy, sparse, and dynamic real-world data? And how must our evaluation methodologies evolve to capture this deeper understanding?*

This section addresses these questions by outlining the key challenges

and promising future directions in GNN research, with a particular focus on their application in recommender systems. We categorize these into four main areas: **theoretical and model-centric challenges** that address the intrinsic limitations of GNN architectures; **data-centric challenges** concerning the quality and characteristics of graph data; **task-specific challenges** related to applying GNNs in complex domains like recommender systems; and **methodological challenges** that call for more rigorous and reproducible evaluation practices.

10.1 Theoretical and model-centric challenges

These challenges are inherent to the architectural and theoretical foundations of GNNs themselves. Addressing them is crucial for building more powerful, robust, and scalable models.

Expressiveness and isomorphism: A well-documented limitation of many GNNs is their inability to distinguish between certain non-isomorphic graph structures. The expressive power of most message-passing GNNs is bounded by the Weisfeiler-Lehman (WL) test of graph isomorphism. This means that if two graphs are indistinguishable by the 1-WL test, a standard GNN will learn the same representation for them, limiting their ability to capture fine-grained topological details (Xu *et al.*, 2019). Research is actively exploring more expressive GNNs, such as those based on higher-order WL tests or subgraph-based message passing. The key challenge is to increase expressive power without incurring prohibitive computational costs, striking a balance between theoretical capability and practical feasibility. This remains an active area of research, with recent surveys mapping the landscape of GNN expressiveness (Zhang *et al.*, 2025a).

Oversmoothing and oversquashing: These two related issues represent a major obstacle to building deep GNNs. **Oversmoothing** is the phenomenon where, as the number of GNN layers increases, the representations of all nodes converge to a single, uninformative point, losing their distinctiveness. **Oversquashing** occurs when information from distant nodes gets "squashed" into a fixed-size vector during message passing, creating an information bottleneck that prevents the model

from effectively capturing long-range dependencies (Topping *et al.*, 2022). Potential solutions include architectural innovations like skip connections, graph rewiring to reduce bottlenecks, and moving beyond the standard message-passing paradigm altogether. Exploring techniques from differential geometry to better understand graph curvature and its impact on information flow is a particularly promising avenue.

Scalability: Applying GNNs to massive, web-scale graphs with billions of nodes and edges remains a major computational hurdle. The "neighborhood explosion" problem, where the number of neighbors to consider grows exponentially with the number of layers, makes full-batch training intractable. While various sampling techniques have been developed, future work should focus on improving their efficiency and reducing bias. Additionally, research into decentralized and parallel training algorithms, as well as hardware-software co-design for GNN acceleration, will be critical for scaling to industrial-sized problems.

10.2 Data-centric challenges

The performance of GNNs is intrinsically tied to the quality and characteristics of the underlying graph data. Real-world graphs, however, present numerous challenges that are often abstracted away in benchmark datasets.

Data scarcity and quality: The field is hampered by a scarcity of large, high-quality, and diverse public datasets. Many real-world graphs are proprietary, forcing researchers to rely on a small set of well-worn benchmarks that may not be representative of real-world complexity (Dwivedi *et al.*, 2023). Furthermore, real-world graphs are often noisy, imbalanced, and contain missing information, which can severely degrade GNN performance (Ju *et al.*, 2026).

Heterophily vs. homophily: A core assumption of many GNNs is *homophily*, that connected nodes are similar. However, many real-world graphs exhibit *heterophily*, where connected nodes are dissimilar. Standard GNNs often fail in such scenarios. While specialized architectures for heterophilous graphs exist, a more unified understanding is needed. Future models should be able to adaptively learn whether to smooth or

sharpen features based on the local graph structure, moving beyond a one-size-fits-all assumption about homophily (Zhu *et al.*, 2020).

Generative models and topology modification: The static nature of most graph datasets is a limitation. The topology of the user-item graph, for instance, could be modified to improve performance or efficiency. A promising direction is the use of recent generative models, such as diffusion models, to leverage and modify the graph topology. These models could potentially reduce the noise and sparsity of the dataset by generating plausible, yet unobserved, edges, thereby enhancing the performance of downstream GNNs. Recent advances have demonstrated the effectiveness of diffusion models across a broad spectrum of graph-based recommendation tasks, including sequential recommendation (Wang *et al.*, 2024d), out-of-distribution generalization (Zhao *et al.*, 2025a; Zhao *et al.*, 2025b), knowledge-aware recommendation (Jiang *et al.*, 2024b; Zheng *et al.*, 2026), graph contrastive learning (Ge *et al.*, 2026; Huang *et al.*, 2026; Jing *et al.*, 2026), and conditional or guided recommendation (Wen *et al.*, 2026; Zhang *et al.*, 2025b; Zhu and Tan, 2026). In a related vein, recent studies are also exploring how Large Language Models (LLMs) can be leveraged to augment or modify the user-item graph topology, further blurring the lines between structured and unstructured data modeling. Beyond the pioneering work of Wei *et al.* (2024b), LLMs have been successfully applied to a variety of graph-based recommendation tasks, including graph reasoning (Sakurai *et al.*, 2025; Wang *et al.*, 2025b; Wang *et al.*, 2024b), graph retrieval-augmented generation for recommendation (Wang *et al.*, 2025a), sequential recommendation (Kim *et al.*, 2026), and multimodal recommendation (Meng *et al.*, 2025). Another emerging research direction exploits LLMs to generate augmented versions of the user-item graph for tasks such as contrastive learning and hard negative sampling, as demonstrated by several recent studies (Li *et al.*, 2026; Luo *et al.*, 2025; Song *et al.*, 2026; Zheng *et al.*, 2025). Despite the growing body of work leveraging diffusion models and LLMs to enhance the performance of GNN-based recommender systems, little attention has been paid to understanding the topological implications of these approaches, with only a few notable exceptions (Fioretti *et al.*, 2025). We argue that investigating these

methods through a topological perspective is crucial for uncovering how they reshape the structure of the user-item graph and, ultimately, why such structural modifications translate into improved recommendation performance.

10.3 Task-specific and application-oriented challenges

As GNNs are applied to more complex tasks, new challenges emerge related to the specific demands of each domain. Recommender systems provide a clear example of this.

Exploiting topology in GNN-based recommendation: While GNNs are a natural fit for recommendation, there is still a limited understanding of how they exploit the topological properties of the user-item graph. Current research often focuses on accuracy, neglecting the rich, interpretable insights that a topological perspective can offer. There is a need for more research that explicitly investigates the link between topological characteristics (i.e., node degree, clustering coefficients, assortativity, degree distribution) and recommendation outcomes like novelty, diversity, and fairness. This involves designing GNNs that are not just powerful predictors but are also tailored to leverage specific, desirable topological properties of the data (Huang *et al.*, 2025; Mo *et al.*, 2025; Zhang *et al.*, 2025c).

Integrating multimodal and sequential data: Recommender systems and other applications often involve rich side information that is not easily captured in a simple graph structure. For example, user behavior is often sequential, and items may have multimodal features (e.g., text, images) (Malitesta *et al.*, 2025). A key challenge is to develop principled ways to integrate this information. This could involve treating multimodal attributes as node features or viewing sequential data as a way of re-structuring the user-item bipartite graph. The goal is to create a unified model that can reason over the graph topology and the rich semantics of its components simultaneously.

10.4 Evaluation and methodological challenges

The way GNN models are evaluated is crucial for driving meaningful progress. Current benchmarking and reproducibility practices, however, face significant criticism.

Benchmarking and reproducibility: The GNN field has been criticized for its lack of standardized and rigorous benchmarking practices, a topic of intense debate within the community (Dong *et al.*, 2025; Dwivedi *et al.*, 2023). Different studies often use different datasets, data splits, and experimental settings, making fair and reproducible comparisons nearly impossible. This can lead to a "reproducibility crisis" where reported gains are difficult to verify. There is an urgent need for the community to rally around high-quality, standardized benchmarks that cover a diverse range of tasks and graph types. This includes not just node classification but also graph classification, link prediction, and applications in science and industry. Platforms that facilitate easy and fair comparison, with open-sourcing code/data, are essential for fostering robust and reproducible research (Bechler-Speicher *et al.*, 2025).

Beyond accuracy: Evaluation often focuses narrowly on a single metric like accuracy, ignoring other crucial aspects of model performance such as fairness, robustness, interpretability, and efficiency. Future work must adopt a more holistic approach to evaluation. This means extending the analysis of GNNs to measures that go beyond accuracy, such as diversity, novelty, bias, and fairness, as it has been done for other families of recommendation algorithms (Deldjoo *et al.*, 2021). A multi-faceted evaluation framework will lead to the development of GNNs that are not just accurate but also reliable, equitable, and trustworthy.